

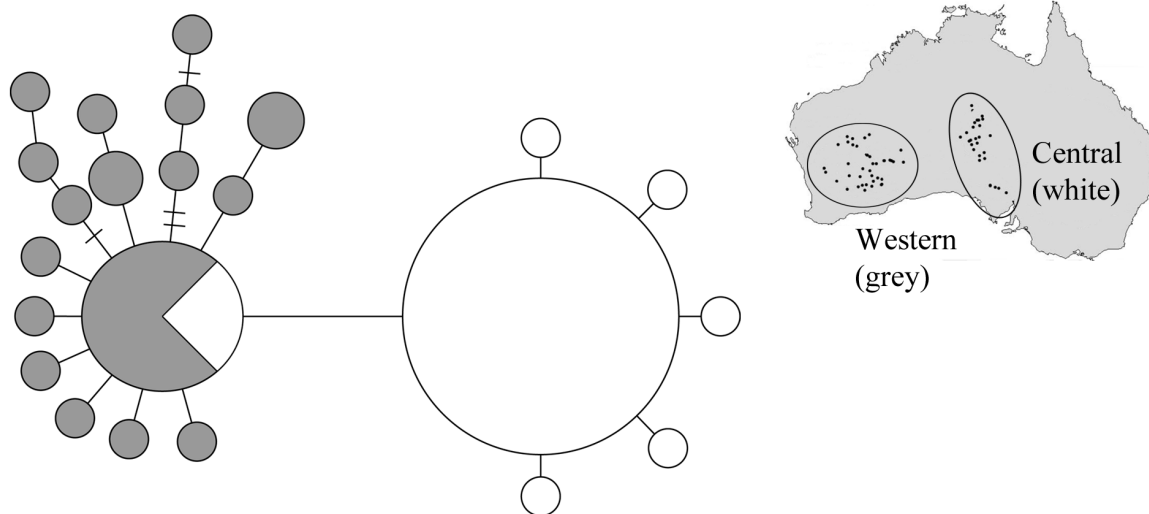
(5 pts) 1. I appreciate your service as a guinea pig during this version of Biogeography. As a way of saying thanks, you get five free points by writing something heartfelt or snarky. Either is fine. Fun is good.

(10 pts) 2. Time to get serious: Describe the difference between the target effect and the rescue effect.

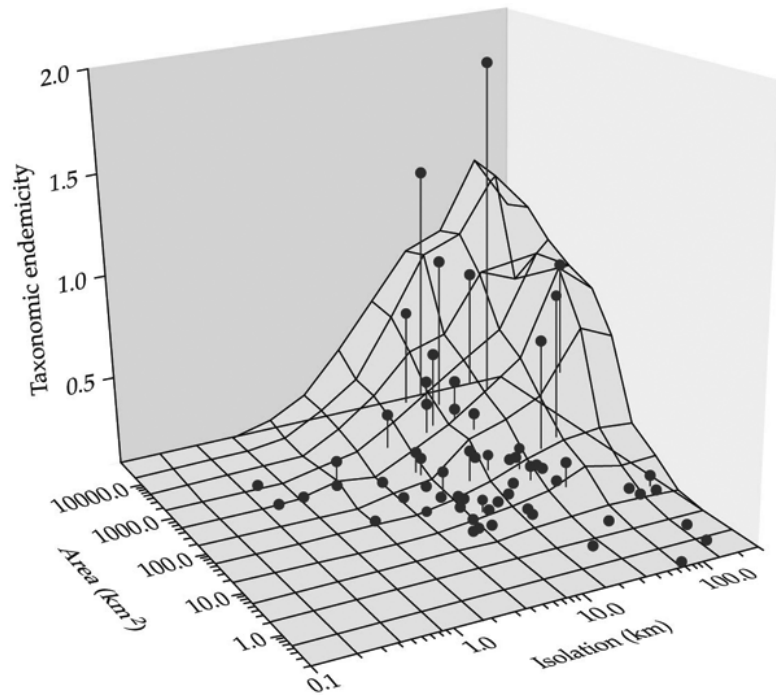
(10 pts) 3. Describe the albedo effect and how it contributed to expansion of Pleistocene glaciers.

- (15 pts) 4. Australia was not covered by glaciers but climate change during the Pleistocene still caused range contraction and expansion for many taxa. An Australian gecko, *Heteronotia binoei*, has the distribution indicated by the black dots in the inset below. The large gap in the distribution suggests one of two scenarios: post-Pleistocene range expansion or a long-term barrier to dispersal between the two regions. Strasburg et al. (2007) tested these alternate hypotheses. Below is a simplified version of their parsimony network. Based on the network, state which hypothesis is supported and describe how the network supports the hypothesis you chose.

Each circle represents one allele. Color corresponds to the population sampled; grey is the western cluster and white is the central cluster. Each line connecting two circles indicates one mutation difference between alleles. Small dashes indicate additional mutational differences between alleles.



- (15 pts) 5. Use Lomolino's extension of the MacArthur-Wilson model of island biogeography to explain why endemism is high on large, isolated islands, as shown in the figure below.



- (20 pts) 6. Abbott's Duiker is a small antelope endemic to the fragmented Eastern Arc forests of Tanzania. Known localities with the duiker are shown in the inset. Assume you are given the task of increasing survivorship and movement of individuals between the southernmost and northernmost locations. Apply whatever island biogeography concepts you think are necessary to accomplish this goal. You must explain *how* each concept you choose helps achieve the goal. You must also discuss how corridors can contribute, but you must also discuss at least one potential disadvantage of corridors (as discussed in class on Tuesday). I have extra paper if you need more writing space.

